



University of Minho
School of Engineering



Aprendizagem Profunda

YOLO (You Only Look Once)

AP @ [MEI,MIA,MECD]/1º ano – 2º Semestre

Diogo Cachetas

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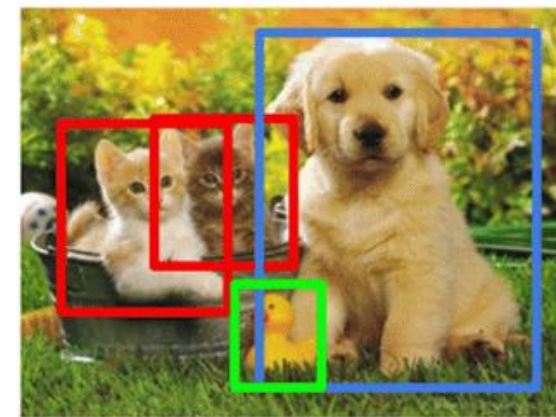
Conclusion



Introduction



CAT



CAT, DOG, DUCK

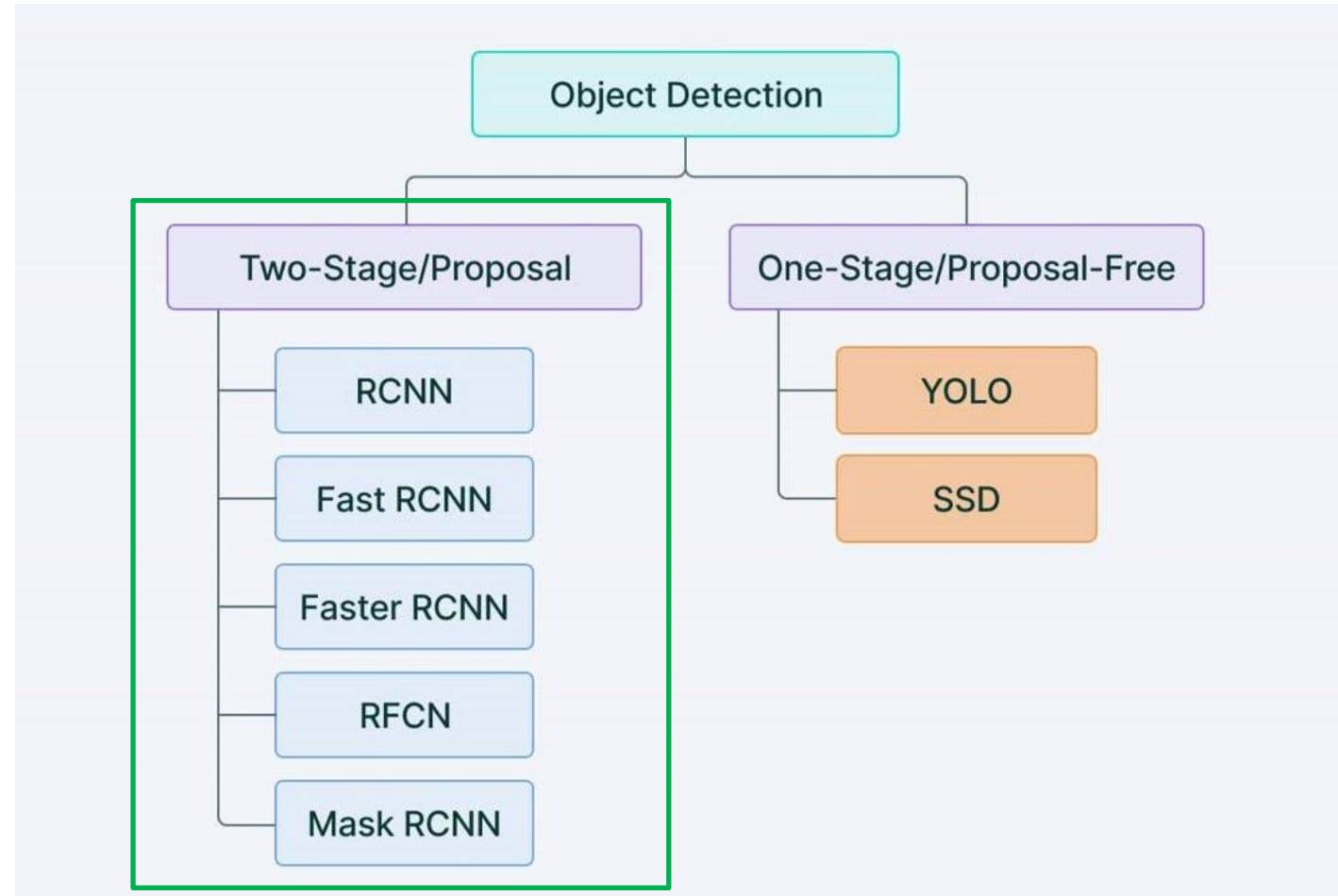
[10.13140/RG.2.2.26822.69443](https://doi.org/10.13140/RG.2.2.26822.69443)

- Unlike image classification, object detection algorithms are capable of identifying and localizing the object of interest in an input image/video by drawing a rectangular bounding box around the targeted object.
- However, compared to image classification task, object detection algorithms require more computational resources for both training and inferencing. Several object detection algorithms were proposed over the past few years achieving very good performances.
- These algorithms can be divided into two principal groups, which are two-stages and single-stage detectors.

Introduction – Two Stage

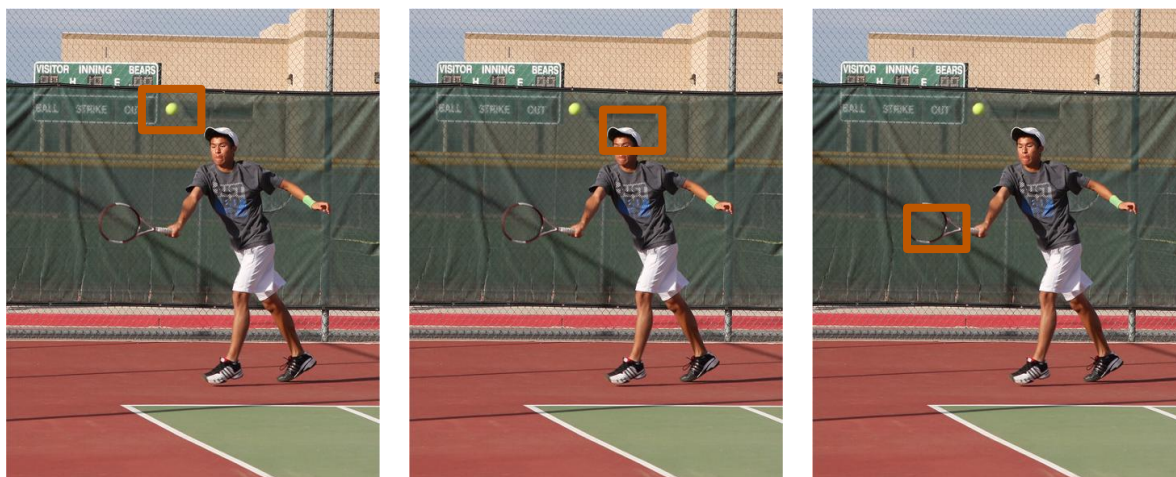
These models are known for high accuracy and precision, making them ideal for tasks where detecting small objects or precise localization is critical.

- **Stage 1: Region Proposal Network (RPN):** Identifies potential regions in the image that might contain an object (Region of Interest - RoI).
- **Stage 2: Classification and Regression:** A separate network classifies the proposed regions and refines the bounding box coordinates.
- **Advantages:** Higher accuracy and better at detecting smaller objects.
- **Disadvantages:** Slower inference speeds, making them less suitable for real-time applications.

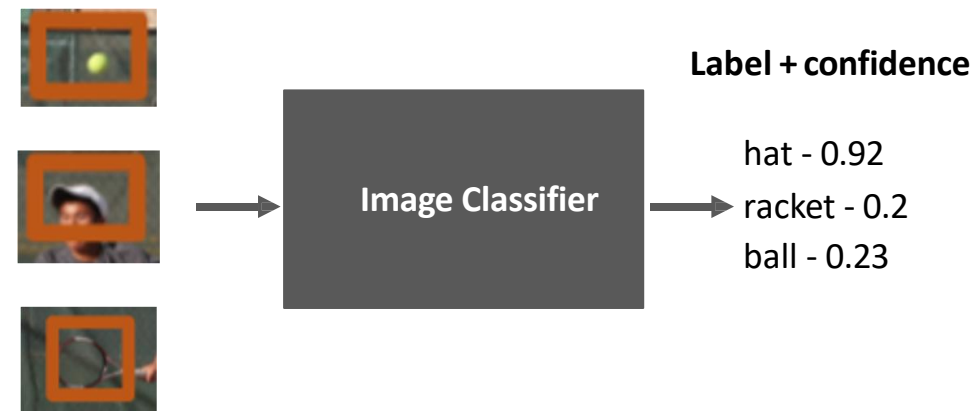


Introduction – Two Stage

This pipeline was used in nearly all SOTA Object Detection prior:



Step 1: Scan the image to generate candidate bounding boxes



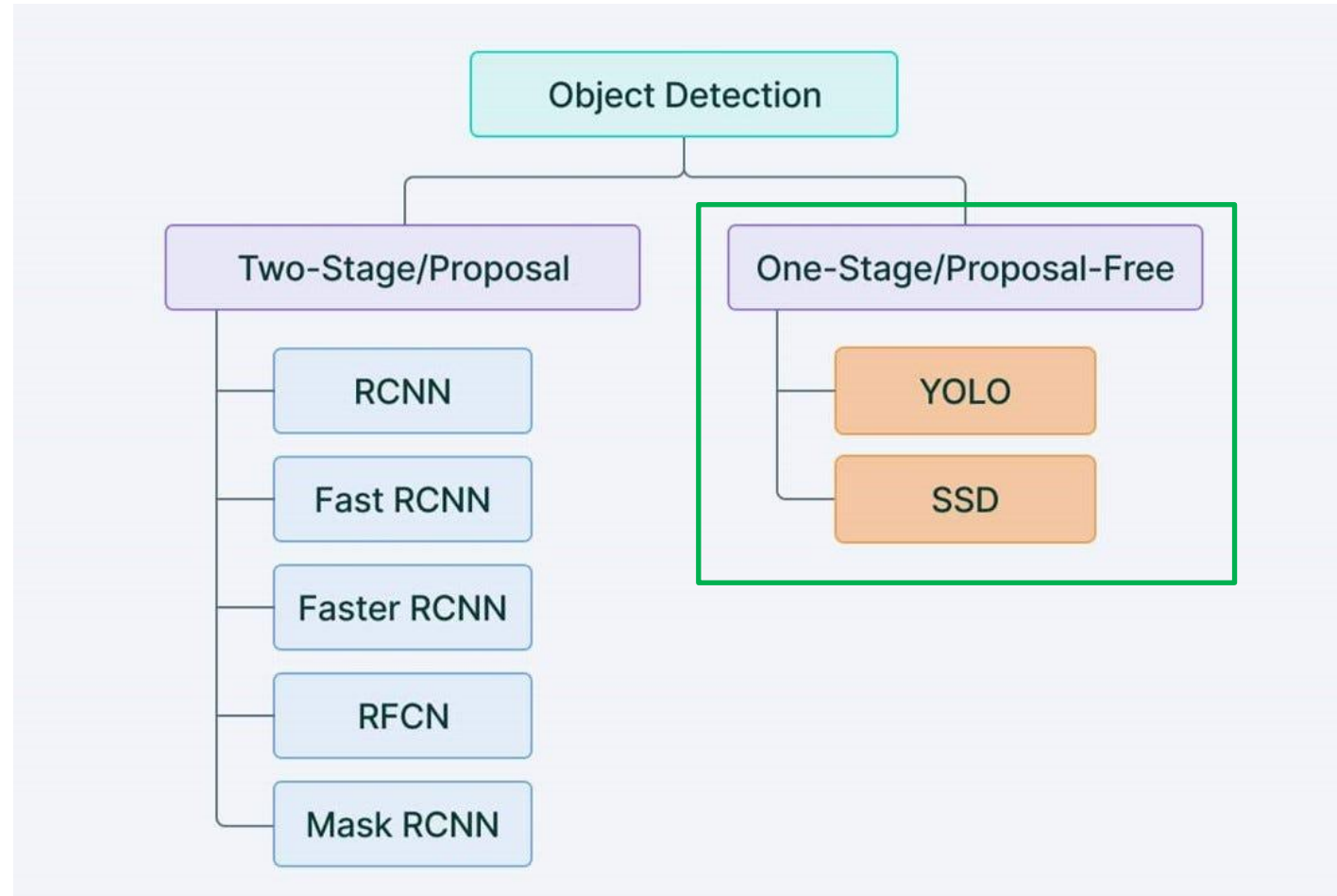
Step 2: Run the bounding box through a classifier

Step 3: Conduct post-processing (filtering out redundant bounding boxes)

Introduction – One Stage

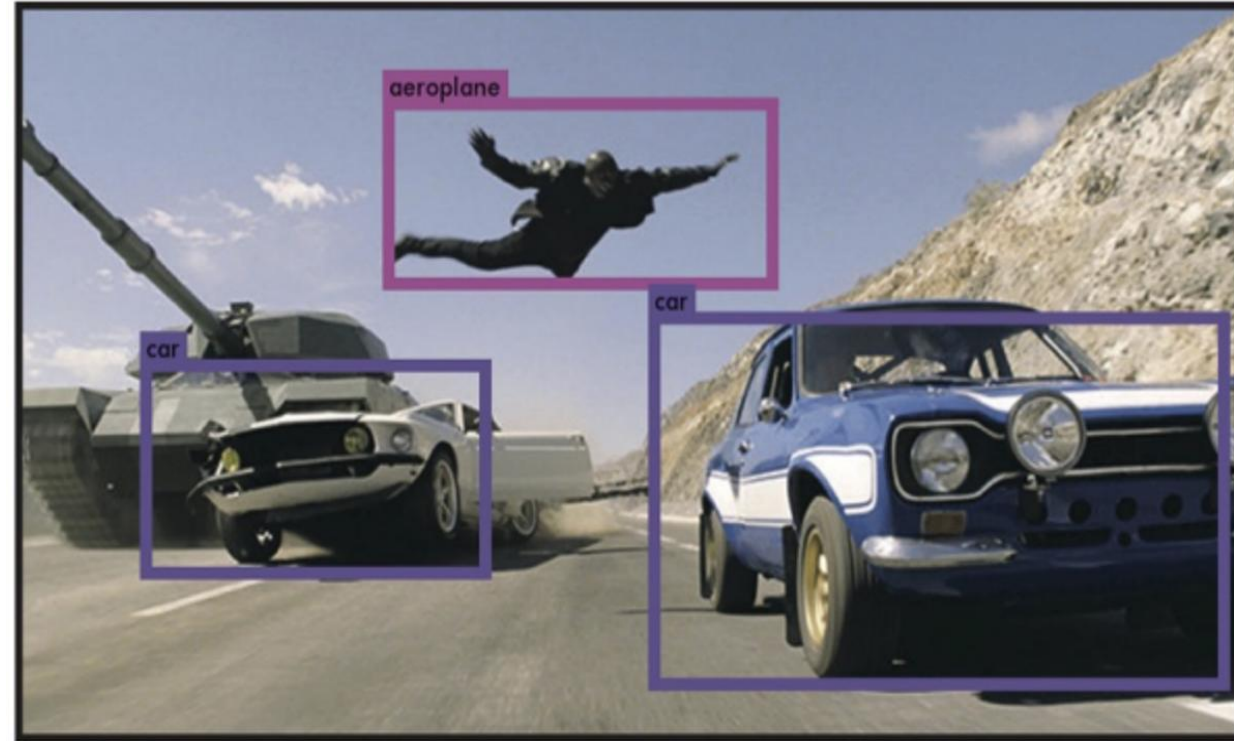
These models are designed for speed and efficiency, treating object detection as a direct regression problem.

- The image passes through a network once, which directly predicts bounding boxes and class probabilities for the entire image simultaneously.
- **Advantages:** Very fast, making them ideal for real-time applications like video analytics, drones, and autonomous driving.
- **Disadvantages:** Generally lower accuracy than two-stage detectors, particularly for small or highly overlapping objects.



Problem Conceptualization

- ❖ Given an image generate bounding boxes, one for each detectable object in image
- ❖ For each bounding box, output 5 predictions.
- ❖ x, y, w, h , confidence and also output class:
 - ❖ x, y (coordinates for center of bounding box)
 - ❖ w, h (width and height)
 - ❖ confidence (probability bounding box has object)
 - ❖ class (classification of object in bounding box)



Introduction – Why YOLO?



Most of other detection systems repurpose classifiers to perform detection.

- DPM (deformable parts model): uses sliding window approach where classifier is run at evenly spaced locations over the entire image.
- R-CNN uses region proposal methods to generate bounding boxes then run a classifier on these proposed boxes.

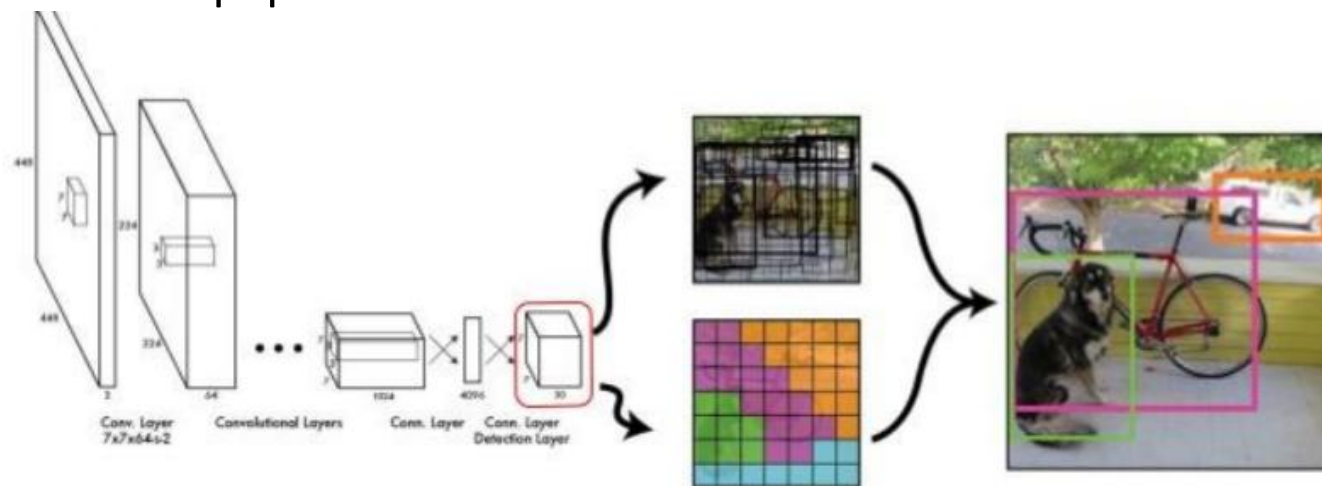
In yolo object detection is reframed as a single regression problem

- Image pixel bounding box coordinate and class probabilities.
- Helps to answer the where and what

Introduction - YOLO

YOLO is an algorithm proposed by Redmond et. al in a research article published at the IEEE/CVF Conference on Computer Vision and pattern Recognition (CVPR) as a conference paper.

- It is a real-time object detection algorithm
- It uses a single CNN to detect objects in images
- It contains three components: **input**, **backbone** and **output**



Input

- Takes an image and applies convolutional layers to extract features

Backbone

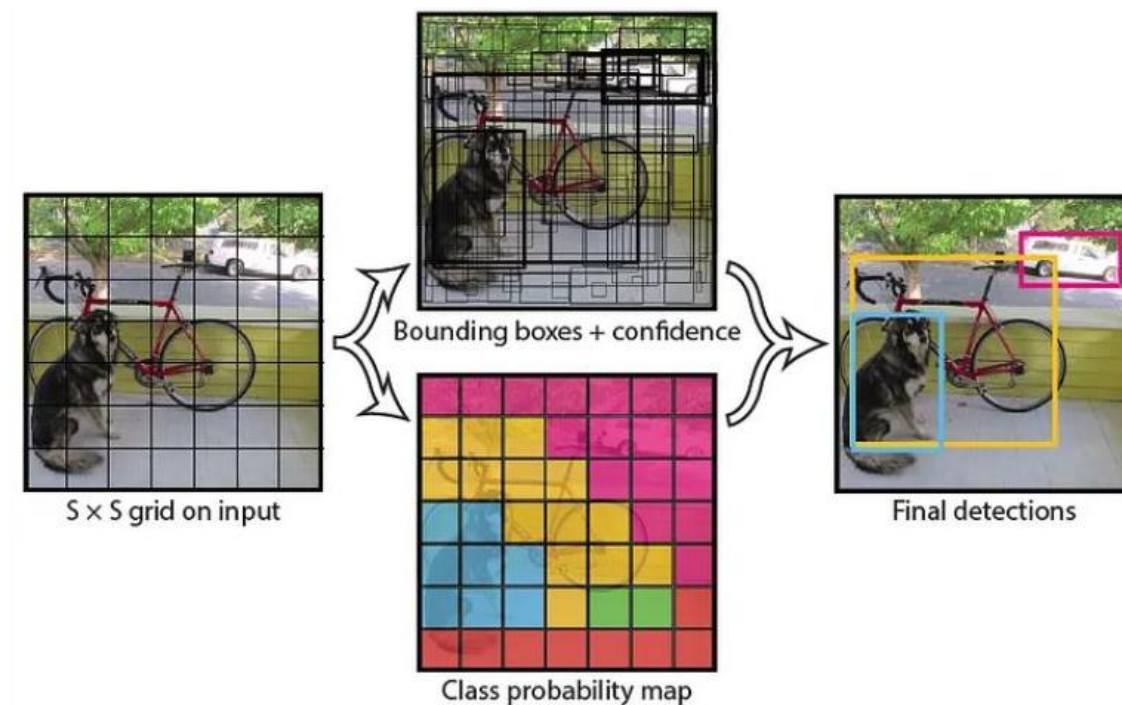
- Contains convolutional and pooling layers that down sample feature maps and extract high-level features

Output

- Predicts bounding boxes, class probabilities, and confidence scores for each detected object

Introduction - YOLO

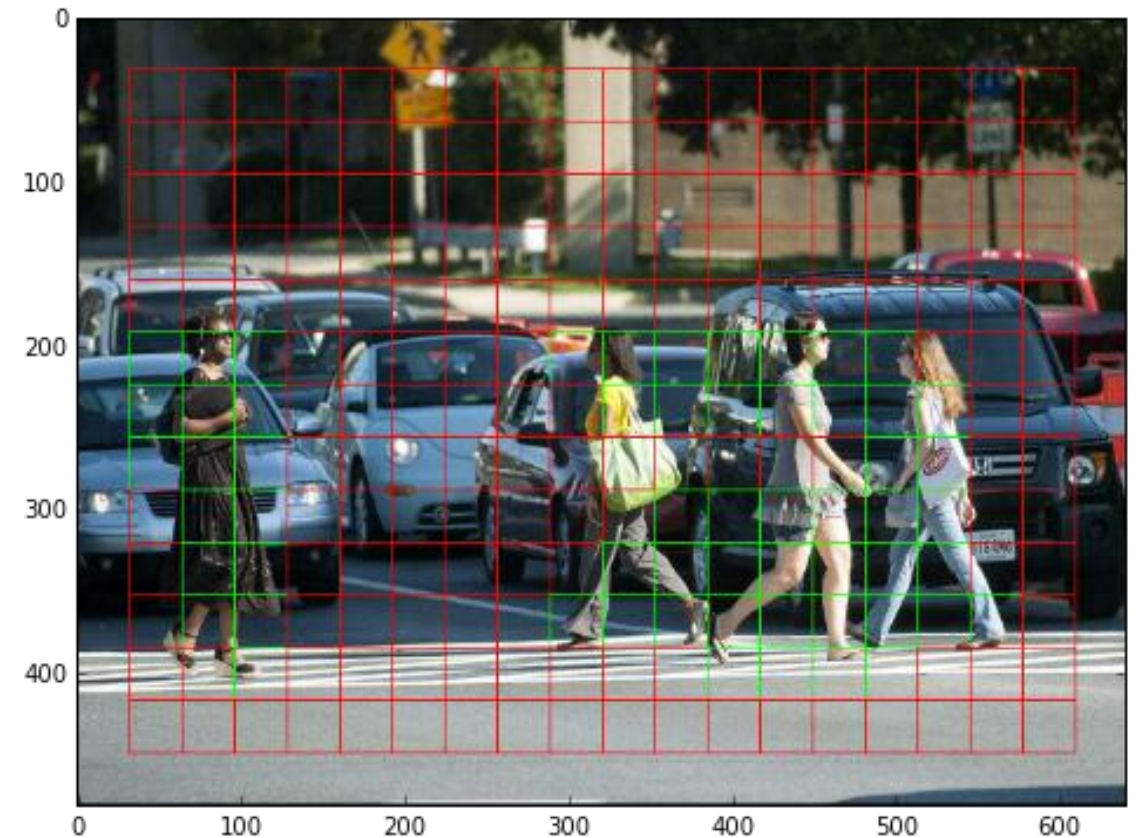
- YOLO is trained on large datasets such as COCO or VOC with labelled images, object classes and bounding boxes.
- During training, YOLO optimizes a loss function that combines localization loss and classification loss
- This results in a model that can accurately detect objects in images
- YOLO is a fast and accurate model, making it popular for real-time applications such as video surveillance, robotics, autonomous vehicles, etc.



YOLO - Architecture

Residual Blocks

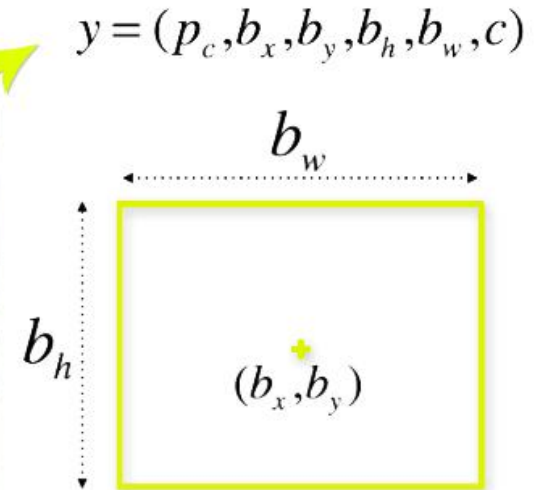
- The image is divided into various grids. Each grid has a dimension of $S \times S$. The following image shows how an input image is divided into grids.
- In the image, there are many grid cells of equal dimension. Every grid cell will detect objects that appear within them.
- Which means if an object center appears within a certain grid cell, then this cell will be responsible for detecting it.



YOLO - Architecture

Bounding box regression

- A bounding box is an outline that highlights an object in an image.
- Every bounding box in the image consists of the following attributes:
 - Width (bw)
 - Height (bh)
 - Class (c) (person, car, traffic light, etc.)
 - Bounding box center (bx,by)



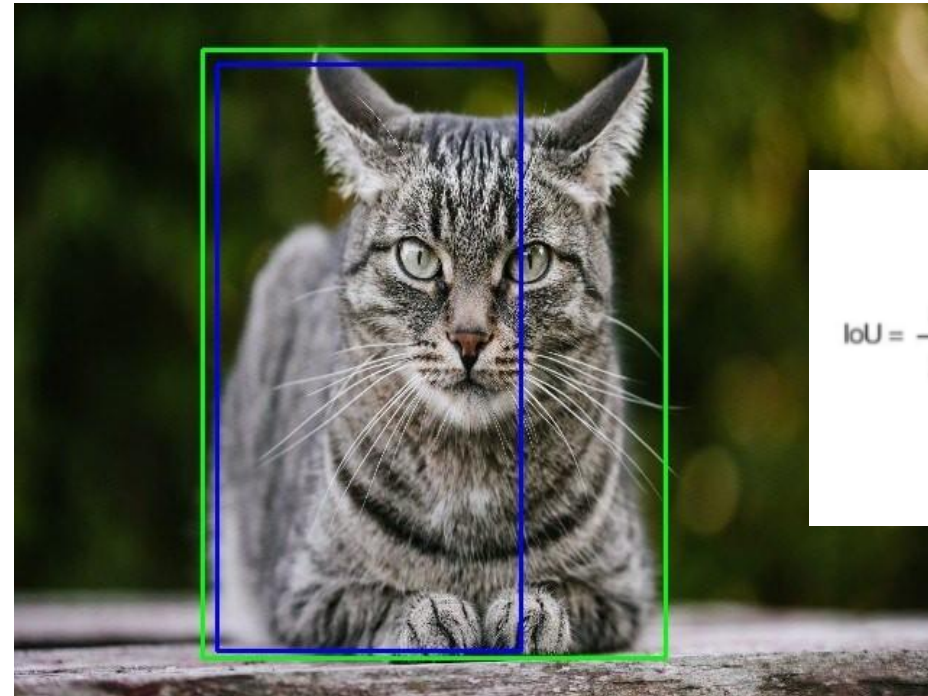
- The image shows an example of a bounding box. The bounding box has been represented by a yellow outline.
- YOLO uses a single bounding box regression to predict the height, width, center, and class of objects. In the image above, represents the probability of an object appearing in the bounding box.

YOLO - Architecture

Intersection over union (IOU)

- Intersection over union (IOU) is a phenomenon in object detection that describes how boxes overlap. YOLO uses IOU to provide an output box that surrounds the objects perfectly.
- Each grid cell is responsible for predicting the bounding boxes and their confidence scores. The IOU is equal to 1 if the predicted bounding box is the same as the real box. This mechanism eliminates bounding boxes that are not equal to the real box.

The following image provides a simple example of how IOU works.



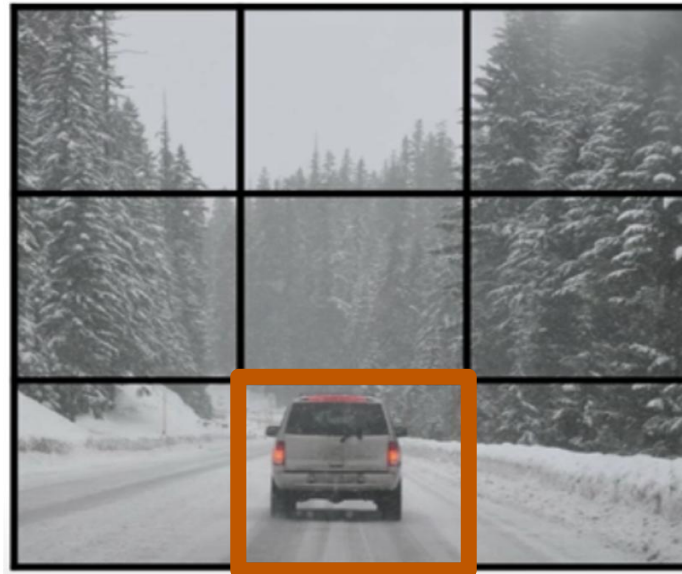
$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

In the image, there are two bounding boxes, one in green and the other one in blue. The blue box is the predicted box while the green box is the real box. YOLO ensures that the two bounding boxes are equal.

YOLO Training

- ❖ YOLO is a regression algorithm. What is X? What is Y?
- ❖ X is simple, just an image width (in pixels) * height (in pixels) * RGB values
- ❖ Y is a tensor of size $S * S * (B * 5 + C)$
- ❖ $B * 5 + C$ term represents the predictions + class predicted distribution for a grid block

For each grid block, we have a vector like this. For this example, B is 2 and C is 2



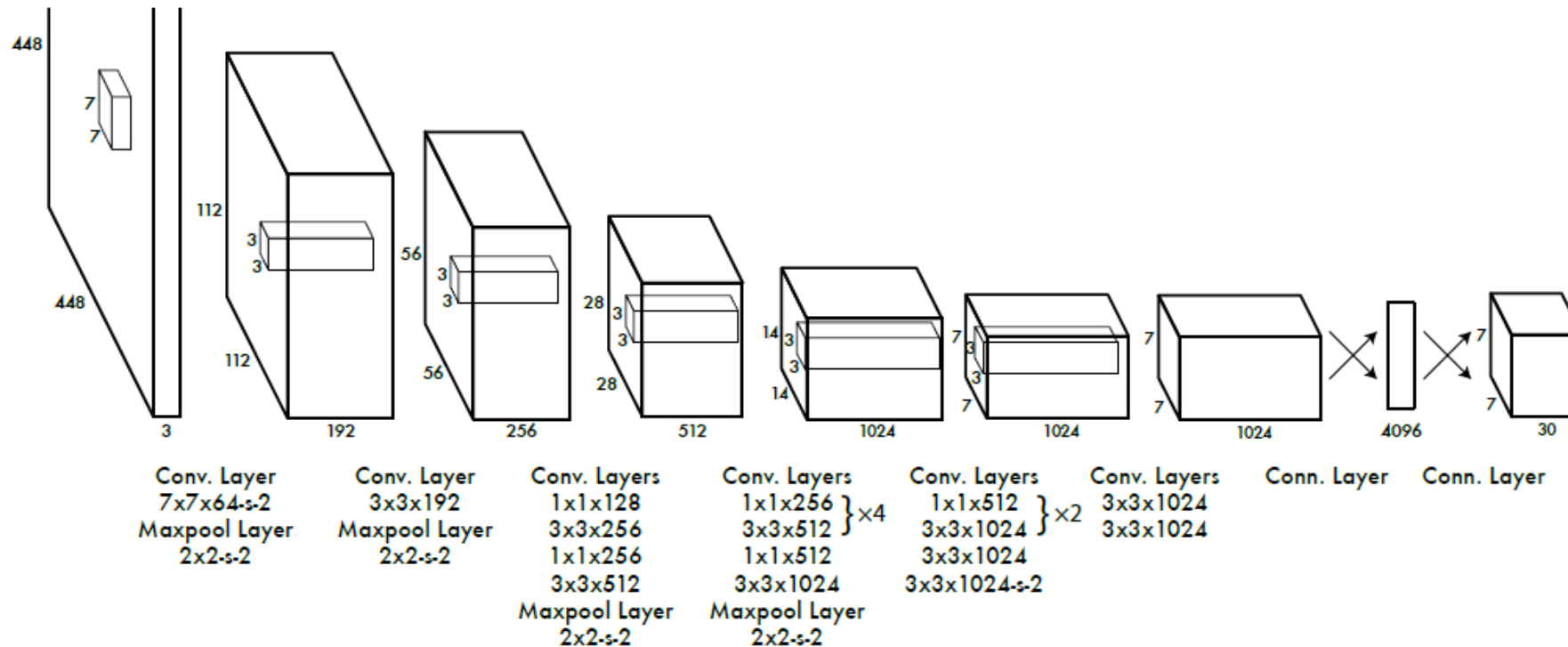
$\mathbf{p}_1$
b_{x_1}
b_{y_1}
b_{h_1}
b_{w_1}
$\mathbf{p}_2$
b_{x_2}
b_{y_2}
b_{h_2}
b_{w_2}
c_1
c_2

GT label
example:

1
b_{x_1}
b_{y_1}
b_{h_1}
b_{w_1}
0
?
?
?
?
$c_1 = 1$
$c_2 = 0$

YOLO – Architecture – All together

24 convolutional layers followed by 2 fully connected layers. Instead of the inception modules used by GoogLeNet, we simply use 1×1 reduction layers followed by 3×3 convolutional layers

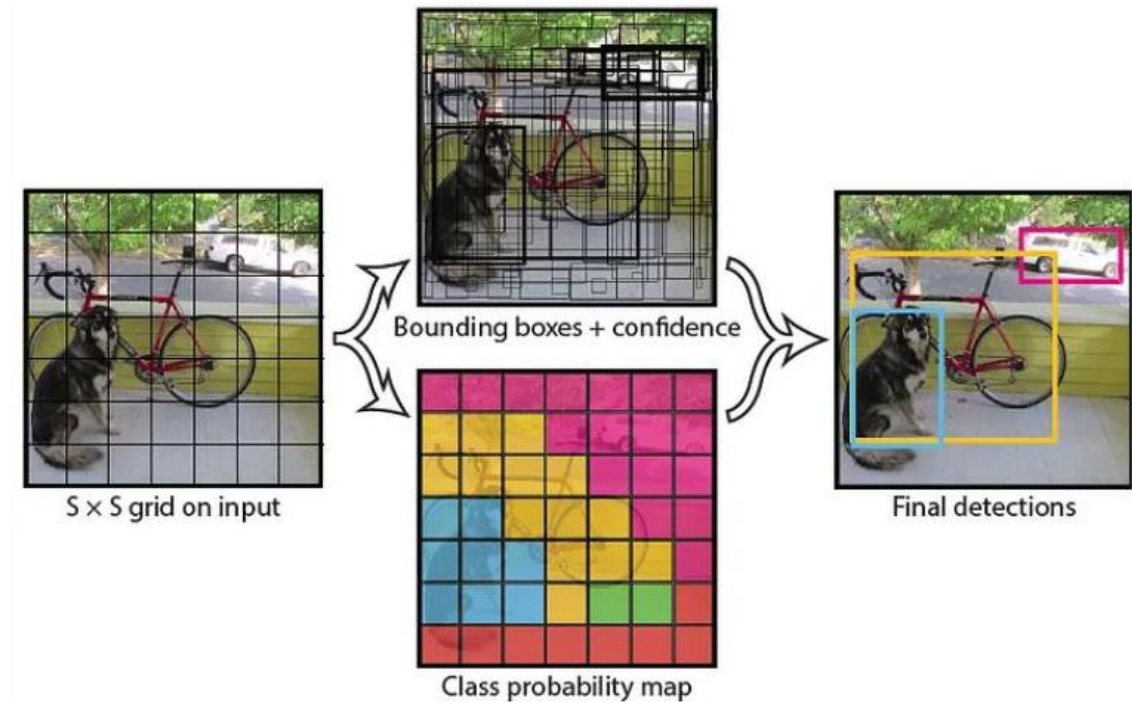


Fast YOLO uses a neural network with fewer convolutional layers (9 instead of 24) and fewer filters in those layers. Other than the size of the network, all training and testing parameters are the same between YOLO and Fast YOLO.

YOLO – Architecture – All together

First, the image is divided into grid cells. Each grid cell forecasts B bounding boxes and provides their confidence scores. The cells predict the class probabilities to establish the class of each object.

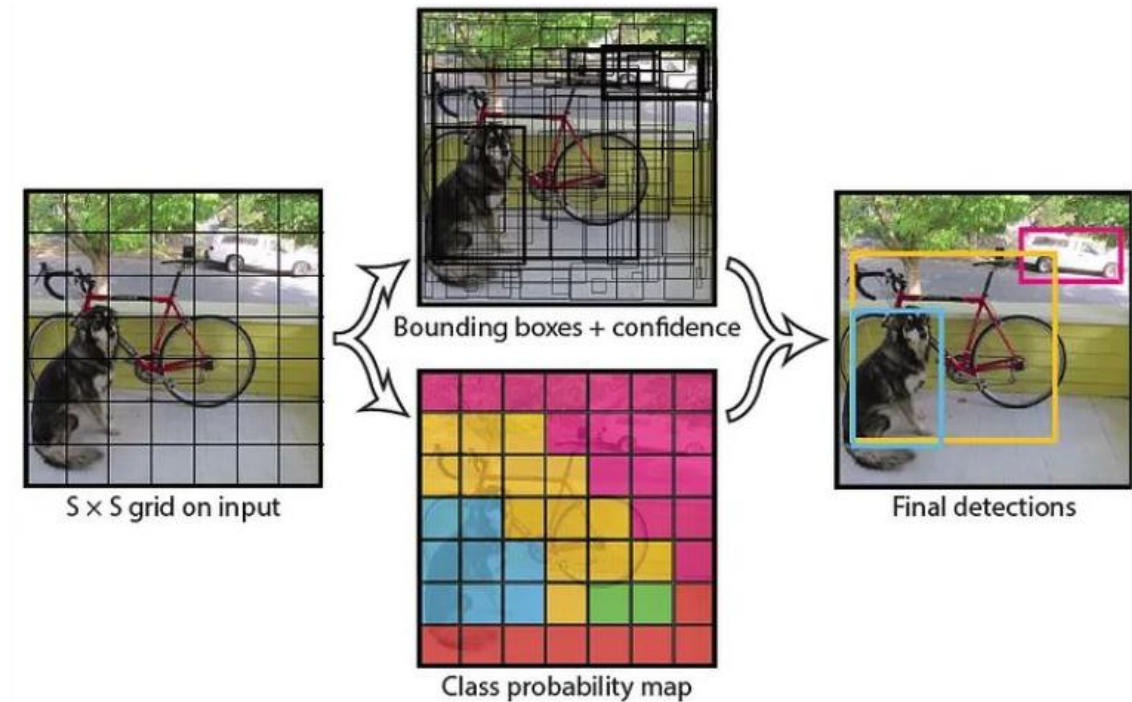
For example, we can notice at least three classes of objects: a car, a dog, and a bicycle. All the predictions are made simultaneously using a single convolutional neural network.



YOLO – Architecture – All together

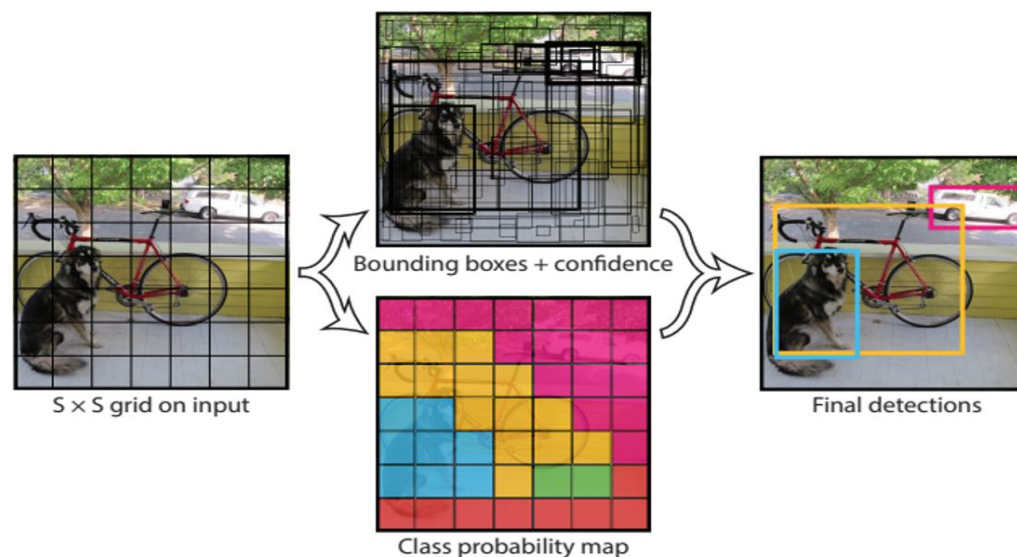
Second, intersection over union ensures that the predicted bounding boxes are equal to the real boxes of the objects. This phenomenon eliminates unnecessary bounding boxes that do not meet the characteristics of the objects (like height and width). The final detection will consist of unique bounding boxes that fit the objects perfectly.

For example, the car is surrounded by the pink bounding box while the bicycle is surrounded by the yellow bounding box. The dog has been highlighted using the blue bounding box.



YOLO Prediction

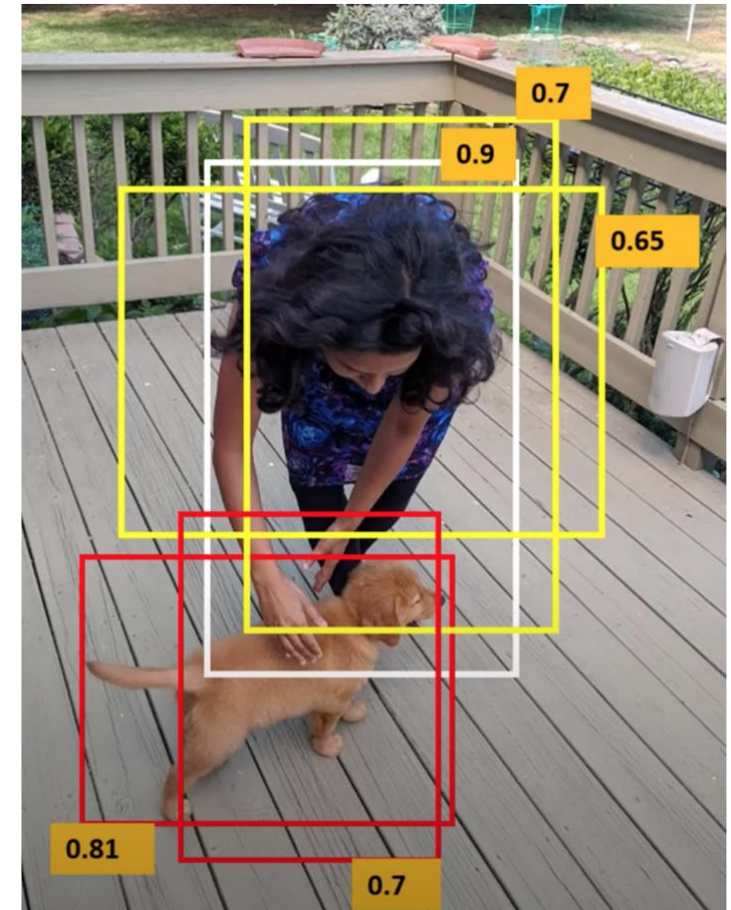
- ❖ We then use the output to make final detections
- ❖ Use a threshold to filter out bounding boxes with low $P(\text{Object})$
- ❖ In order to know the class for the bounding box compute score take argmax over the distribution $\text{Pr}(\text{Class}|\text{Object})$ for the grid the bounding box's center is in



$$\text{Pr}(\text{Class}_i|\text{Object}) * \text{Pr}(\text{Object}) * \text{IOU}_{\text{pred}}^{\text{truth}} = \text{Pr}(\text{Class}_i) * \text{IOU}_{\text{pred}}^{\text{truth}}$$

NMS – Non-maximal suppression

- ❖ Most of the time objects fall in one grid, however it is still possible to get redundant boxes (rare case as object must be close to multiple grid cells for this to happen)
- ❖ Discard bounding box with high overlap (keeping the bounding box with highest confidence)
- ❖ Adds 2-3% on final mAP score



YOLO – Objective Function

- ❖ For YOLO, we need to minimize the following loss functions
- ❖ Sum squared error is used

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\ + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right]$$

Coordinate Loss: Minimize the difference between x,y,w,h pred and x,y,w,h ground truth. ONLY IF object exists in grid box and if bounding box is resp for pred

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2$$

Confidence Loss: Loss based on confidence ONLY IF there is object

$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2$$

No Object Loss: based on confidence if there is no object

$$+ \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2$$

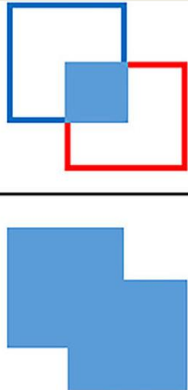
Class loss: minimize loss between true class of object in grid box



Application Study

Benchmark - Setup

- ❖ In this work, the authors compare YOLO against the previous work described on PASCAL VOC 2007, and VOC 2012 as well as out of domain art dataset
- ❖ Correct if IOU metric above 0.5 and class is correct
- ❖ Use two performance metrics:
 - mAP score: mean average precision
 - FPS: frames per second

$$IoU = \frac{\text{area of intersection}}{\text{are of union}} =$$


Benchmark - Results

- ❖ Baseline YOLO outperform real time detectors by a large amount
- ❖ Performs better than most and requires less time as well

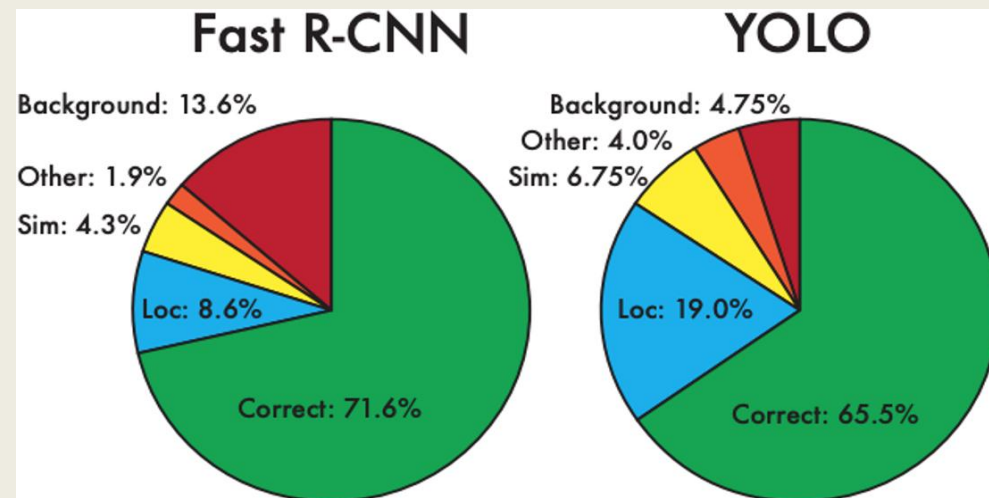
Real-Time Detectors	Train	mAP	FPS
100Hz DPM [31]	2007	16.0	100
30Hz DPM [31]	2007	26.1	30
Fast YOLO	2007+2012	52.7	155
YOLO	2007+2012	63.4	45
Less Than Real-Time			
Fastest DPM [38]	2007	30.4	15
R-CNN Minus R [20]	2007	53.5	6
Fast R-CNN [14]	2007+2012	70.0	0.5
Faster R-CNN VGG-16[28]	2007+2012	73.2	7
Faster R-CNN ZF [28]	2007+2012	62.1	18
YOLO VGG-16	2007+2012	66.4	21

Benchmark - Results

VOC 2012 test	mAP	aero	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	mbike	person	plant	sheep	sofa	train	tv
MR_CNN_MORE_DATA [11]	73.9	85.5	82.9	76.6	57.8	62.7	79.4	77.2	86.6	55.0	79.1	62.2	87.0	83.4	84.7	78.9	45.3	73.4	65.8	80.3	74.0
HyperNet_VGG	71.4	84.2	78.5	73.6	55.6	53.7	78.7	79.8	87.7	49.6	74.9	52.1	86.0	81.7	83.3	81.8	48.6	73.5	59.4	79.9	65.7
HyperNet_SP	71.3	84.1	78.3	73.3	55.5	53.6	78.6	79.6	87.5	49.5	74.9	52.1	85.6	81.6	83.2	81.6	48.4	73.2	59.3	79.7	65.6
Fast R-CNN + YOLO	70.7	83.4	78.5	73.5	55.8	43.4	79.1	73.1	89.4	49.4	75.5	57.0	87.5	80.9	81.0	74.7	41.8	71.5	68.5	82.1	67.2
MR_CNN_S_CNN [11]	70.7	85.0	79.6	71.5	55.3	57.7	76.0	73.9	84.6	50.5	74.3	61.7	85.5	79.9	81.7	76.4	41.0	69.0	61.2	77.7	72.1
Faster R-CNN [28]	70.4	84.9	79.8	74.3	53.9	49.8	77.5	75.9	88.5	45.6	77.1	55.3	86.9	81.7	80.9	79.6	40.1	72.6	60.9	81.2	61.5
DEEP_ENS_COCO	70.1	84.0	79.4	71.6	51.9	51.1	74.1	72.1	88.6	48.3	73.4	57.8	86.1	80.0	80.7	70.4	46.6	69.6	68.8	75.9	71.4
NoC [29]	68.8	82.8	79.0	71.6	52.3	53.7	74.1	69.0	84.9	46.9	74.3	53.1	85.0	81.3	79.5	72.2	38.9	72.4	59.5	76.7	68.1
Fast R-CNN [14]	68.4	82.3	78.4	70.8	52.3	38.7	77.8	71.6	89.3	44.2	73.0	55.0	87.5	80.5	80.8	72.0	35.1	68.3	65.7	80.4	64.2
UMICH_FGS_STRUCT	66.4	82.9	76.1	64.1	44.6	49.4	70.3	71.2	84.6	42.7	68.6	55.8	82.7	77.1	79.9	68.7	41.4	69.0	60.0	72.0	66.2
NUS_NIN_C2000 [7]	63.8	80.2	73.8	61.9	43.7	43.0	70.3	67.6	80.7	41.9	69.7	51.7	78.2	75.2	76.9	65.1	38.6	68.3	58.0	68.7	63.3
BabyLearning [7]	63.2	78.0	74.2	61.3	45.7	42.7	68.2	66.8	80.2	40.6	70.0	49.8	79.0	74.5	77.9	64.0	35.3	67.9	55.7	68.7	62.6
NUS_NIN	62.4	77.9	73.1	62.6	39.5	43.3	69.1	66.4	78.9	39.1	68.1	50.0	77.2	71.3	76.1	64.7	38.4	66.9	56.2	66.9	62.7
R-CNN VGG BB [13]	62.4	79.6	72.7	61.9	41.2	41.9	65.9	66.4	84.6	38.5	67.2	46.7	82.0	74.8	76.0	65.2	35.6	65.4	54.2	67.4	60.3
R-CNN VGG [13]	59.2	76.8	70.9	56.6	37.5	36.9	62.9	63.6	81.1	35.7	64.3	43.9	80.4	71.6	74.0	60.0	30.8	63.4	52.0	63.5	58.7
YOLO	57.9	77.0	67.2	57.7	38.3	22.7	68.3	55.9	81.4	36.2	60.8	48.5	77.2	72.3	71.3	63.5	28.9	52.2	54.8	73.9	50.8
Feature Edit [33]	56.3	74.6	69.1	54.4	39.1	33.1	65.2	62.7	69.7	30.8	56.0	44.6	70.0	64.4	71.1	60.2	33.3	61.3	46.4	61.7	57.8
R-CNN BB [13]	53.3	71.8	65.8	52.0	34.1	32.6	59.6	60.0	69.8	27.6	52.0	41.7	69.6	61.3	68.3	57.8	29.6	57.8	40.9	59.3	54.1
SDS [16]	50.7	69.7	58.4	48.5	28.3	28.8	61.3	57.5	70.8	24.1	50.7	35.9	64.9	59.1	65.8	57.1	26.0	58.8	38.6	58.9	50.7
R-CNN [13]	49.6	68.1	63.8	46.1	29.4	27.9	56.6	57.0	65.9	26.5	48.7	39.5	66.2	57.3	65.4	53.2	26.2	54.5	38.1	50.6	51.6

Results - Error Analysis

- Makes far less background errors (less likely to predict false positives on background)
 - IOU is VERY small with any ground truth label
- But far more localization errors
- Correct class, IOU is somewhat small



Localization error



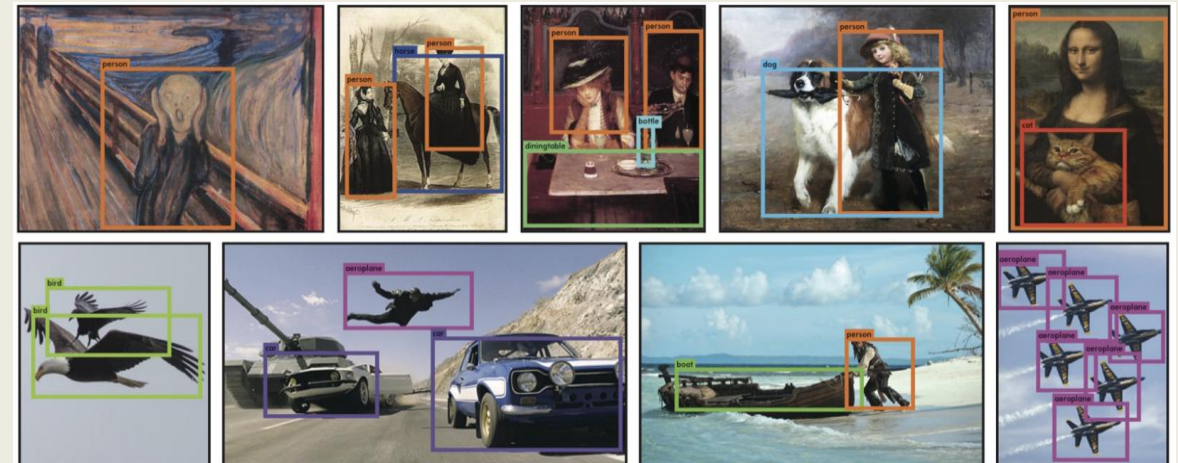
Background error



Results - Out of Domain

- ❖ Ran YOLO + competitors (trained on natural images) on art
- ❖ Does well on artistic datasets where having more global context greatly helps

	VOC 2007	Picasso		People-Art
	AP	AP	Best F_1	AP
YOLO	59.2	53.3	0.590	45
R-CNN	54.2	10.4	0.226	26
DPM	43.2	37.8	0.458	32
Poselets [2]	36.5	17.8	0.271	
D&T [4]	-	1.9	0.051	



YOLO



Limitations of YOLO

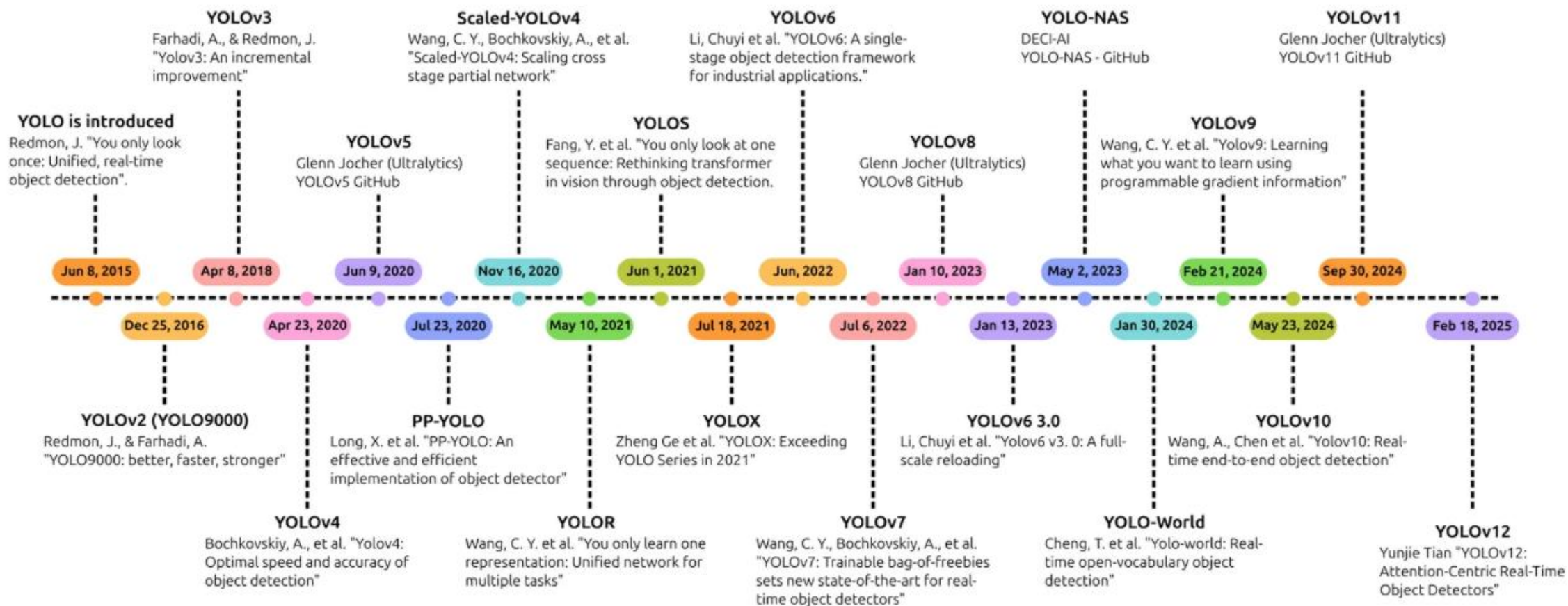
Despite its strengths, YOLO struggles with smaller objects and overlapping detections due to its grid-based division. This can lead to missed detections and reduced accuracy in complex scenarios compared to region-based models.

Advantages of YOLO

YOLO's primary advantage is its speed; it processes images in real-time, making it suitable for applications requiring instant decision-making. Additionally, it comprehensively detects multiple objects within a single model, enhancing overall functionality.

YOLO – Timeline

<https://doi.org/10.1038/s41598-025-26645-2>



YOLO evolution timeline (2015–2025). The timeline illustrates the chronological development of the YOLO object detection framework, from the original YOLOv1 introduced in 2015 to recent versions like YOLOv8 and YOLO-NAS. Each major version shows advancements in architecture, speed, and accuracy, with significant improvements in feature extraction, object localization, and real-time performance.

YOLO – Applications

[nature](#) > [scientific reports](#) > [articles](#) > [article](#)

Article | [Open access](#) | Published: 07 November 2025

Analyzing the enhancement of CNN-YOLO and transformer based architectures for real-time animal detection in complex ecological environments

[Ali Raza](#) , [Fareeha Hanif](#) & [Heba Abdelgader Mohammed](#) 

[Scientific Reports](#)



Contents lists available at [ScienceDirect](#)

Signal Processing

journal homepage: www.elsevier.com/locate/sigpro



A review on early wildfire detection from unmanned aerial vehicles using deep learning-based computer vision algorithms

[Abdelmalek Bouguettaya](#)^a, [Hafed Zarzour](#)^{b,*}, [Amine Mohammed Taberkit](#)^a, [Ahmed Kechida](#)^a

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Remote Sensing Applications: Society and Environment

Volume 42, April 2026, 102059



YOLO-FR: An enhanced object detector for small object detection in complex scenes

[Rui Fan](#)^{a,b}, [Renhao Jiao](#)^{a,b}, [Weigui Nan](#)^{a,b}, [Jin Dang](#)^{a,b}, [Zhixue Wang](#)^{a,b}, [Xuefeng Ge](#)^{a,b}, [Baiying Dong](#)^{a,b} , [Xiaoli Luo](#)^{a,b} 



Biomedical Signal Processing and Control

Volume 113, Part A, March 2026, 108696

YOLO-BT: A novel brain tumor detector with gradient-guided feature diffusion

[Dongwei Zhuang](#) , [Qingan Yao](#) , [Yuncong Feng](#) , [Han Xv](#) , [Congmin Zhang](#) , [Xuexiao Wang](#) 



Procedia Computer Science

Volume 246, 2024, Pages 2792-2801



Real-time Object Detection in Autonomous Vehicles with YOLO

[Nusaybah M. Alahdal](#)^a , [Felwa Abukhodair](#)^{a,b} , [Leila Haj Me](#)



Neurocomputing

Volume 681, 7 June 2026, 133328



SDF-YOLO: A multi-scale dynamic feature fusion network for enhanced lightweight object detection in autonomous driving

[Lei Lu](#)^a , [Longchun Wang](#)^{b,d} , [Jie Ren](#)^c , [Dongming Li](#)^b 



Biomedical Signal Processing and Control

Volume 115, 15 April 2026, 109410

